

1.2 Classwork: Arithmetic Sequences with Variables — Markscheme

1. [Maximum mark: 7]

The sequence 83, 77, 71, 65, ...

(a) $d = 77 - 83 = -6$ A1

The negative sign is required: $d = 6$ earns no mark.

(b) $u_n = u_1 + (n - 1)d = 83 + (n - 1)(-6)$ M1

$u_n = 83 - 6(n - 1)$ A1

Accept the simplified form $u_n = 89 - 6n$, or any other equivalent expression, for full marks. The substituted form is preferred because it shows where the expression comes from.

FT: award M1A1 for a correct expression formed from $u_1 = 83$ and the candidate's own d from (a).

(c) $u_{20} = 83 - 6(20 - 1) = 83 - 114 = -31$ A1

FT: award A1 for the value obtained by substituting $n = 20$ into the candidate's own expression from (b).

(d) $83 - 6(n - 1) = -13$ M1

$-6(n - 1) = -96$

$n - 1 = 16$ A1

$n = 17$ A1

Accept the equivalent set-up $89 - 6n = -13$ leading to $6n = 102$.

The A1 for $n - 1 = 16$ is for any correct intermediate line that isolates n (e.g. $6n = 102$); it does not have to be written in this exact form.

An equation that drops the $(n - 1)$, such as $83 - 6n = -13$ (giving $n = 16$), earns the M1 only — unless $83 - 6n$ is the candidate's own expression from (b), in which case FT applies.

A candidate who states $n = 17$ from a GDC table with no algebraic working earns the final A1 only, since the question asks for algebraic working.

FT: award the marks for a correct solution of the candidate's own expression from (b) set equal to -13 .

2. [Maximum mark: 7]

$$u_1 = k - 4, \quad u_2 = 2k, \quad u_3 = 4k - 1.$$

$$(a) \quad u_2 - u_1 = 2k - (k - 4) = k + 4 \quad \text{A1}$$

$$u_3 - u_2 = (4k - 1) - 2k = 2k - 1 \quad \text{A1}$$

A1 for each expression, awarded independently. Accept a correct unsimplified form, e.g. $2k - k + 4$.

A sign error when subtracting the bracket, e.g. $2k - k - 4 = k - 4$, earns no mark for that expression.

(b) The sequence is arithmetic, so the two differences are equal.

$$k + 4 = 2k - 1 \quad \text{M1}$$

$$k = 5 \quad \text{A1}$$

M1 is for any valid use of the equal-difference property: equating the two differences, using $2u_2 = u_1 + u_3$ (so $4k = 5k - 5$), or substituting a value of k and showing that the two differences are then equal.

FT: award M1A1 for correctly solving the equation formed from the candidate's own expressions in (a).

$$(c) \quad u_1 = 1, \quad u_2 = 10, \quad u_3 = 19 \quad \text{A1}$$

All three terms required for the mark.

FT: award A1 for the three terms obtained by substituting the candidate's own k from (b).

$$(d) \quad d = 10 - 1 = 9$$

$$u_{10} = 1 + (10 - 1)(9) \quad \text{M1}$$

$$u_{10} = 82 \quad \text{A1}$$

M1 is for any correct use of $u_n = u_1 + (n - 1)d$ with the candidate's first term and common difference, or for continuing the list of terms to the 10th term.

Using 10 in place of $(n - 1)$, giving $1 + 10(9) = 91$, earns the M1 only.

FT: award M1A1 for the 10th term correctly obtained from the candidate's own terms in (c).